Member 5

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Chapter 6

Testing Framework

This chapter presents the comprehensive testing strategy implemented to validate the performance, reliability, and robustness of **ReqMind AI**. The testing phase ensures that the system accurately converts raw requirement text into structured outputs while handling errors, edge cases, and unexpected inputs effectively.

The testing process includes functional testing, edge case testing, API validation, JSON structure validation, and export functionality testing.

6.1 Testing Scenarios and Validation Result

Scenario	Input	Expected output	Actual output	Status
Empty input	""	Error message: “Please enter requirements.”	Error displayed correctly	Completed
Simple Requirement	"User login system"	Functional requirement extracted in structured JSON	Functional requirement generated correctly	Completed
Complex SRS	5,000-word document	Full structured output with functional & non-functional requirements	Structured output generated (response time slightly high)	Completed
Multilingual Input	Urdu requirement text	Structured output generated	Output generated (minor formatting adjustments needed)	Completed

Ambiguous Requirement	"System should be fast"	Ambiguity highlighted + clarification question generated	Ambiguity detected & question generated	Completed
Missing Details	"Build an app for students"	Clarification questions generated	Questions generated correctly	Completed
API Failure Simulation	Invalid API token	Proper API error message displayed	Error handled with message	Completed
Invalid JSON Format	Corrupted AI response	JSON error handled gracefully	Error caught via try-except	Completed
Special Characters	Emojis & symbols	Clean parsing without crash	Clean output generated	Completed
PDF Export	Valid structured output	PDF generated & downloadable	PDF downloaded successfully	Completed
Upload file	File	Functional requirement extracted in structured JSON	Functional requirement extracted in structured JSON	Completed

6.2 Error Handling Strategy

To ensure system stability, the following error-handling mechanisms were implemented:

1. Empty Input Validation

If the user submits without entering text, the system displays:

```
Please paste requirements OR upload a file.
```

2. API Error Handling

In case of network failure, token error, or rate limit:

System catches exception using try-except

Displays user-friendly message:

"AI service temporarily unavailable. Please try again."

3. JSON Validation

AI output is validated before processing:

If JSON format is incorrect

System shows:

"Unexpected AI response format. Please retry."

4. Output Cleaning

- Extra spaces removed
- Missing keys replaced with default values

Ensured consistent structure:

a) Functional Requirements
b) Non-Functional Requirements
c) Ambiguities
d) Clarification Questions

5. Performance Testing

- Large document tested (5,000+ words)
- Response time monitored
- System remained stable without crashing

Chapter 7

Expected Outcomes

The implementation of ReqMind AI is expected to significantly enhance the requirement engineering process by automating analysis and improving clarity.

Key Outcomes:

- Reduced requirement analysis time

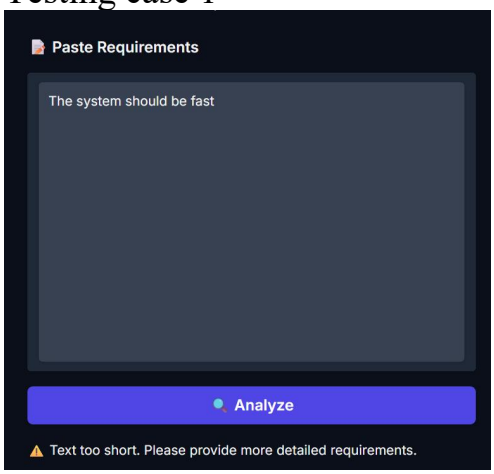
- Improved requirement documentation clarity
- Early detection of ambiguities
- Reduced scope creep
- Improved project planning accuracy
- Better communication between stakeholders
- Structured and standardized requirement outputs
- Increased productivity for development teams

Conclusion :

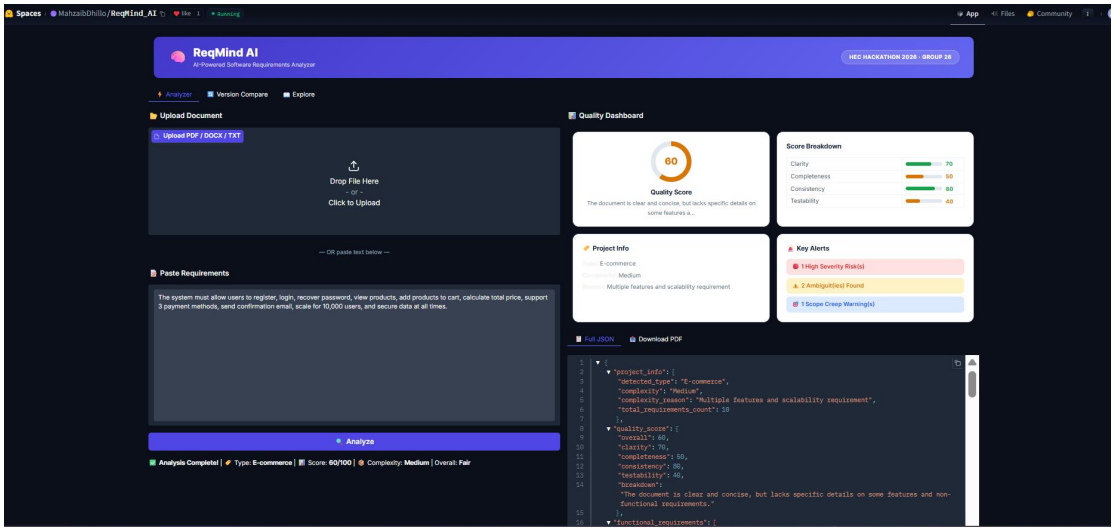
The testing phase ensured robustness under normal, edge, and failure conditions, demonstrating that ReqMind AI is production-ready and scalable for real-world software requirement engineering environments.

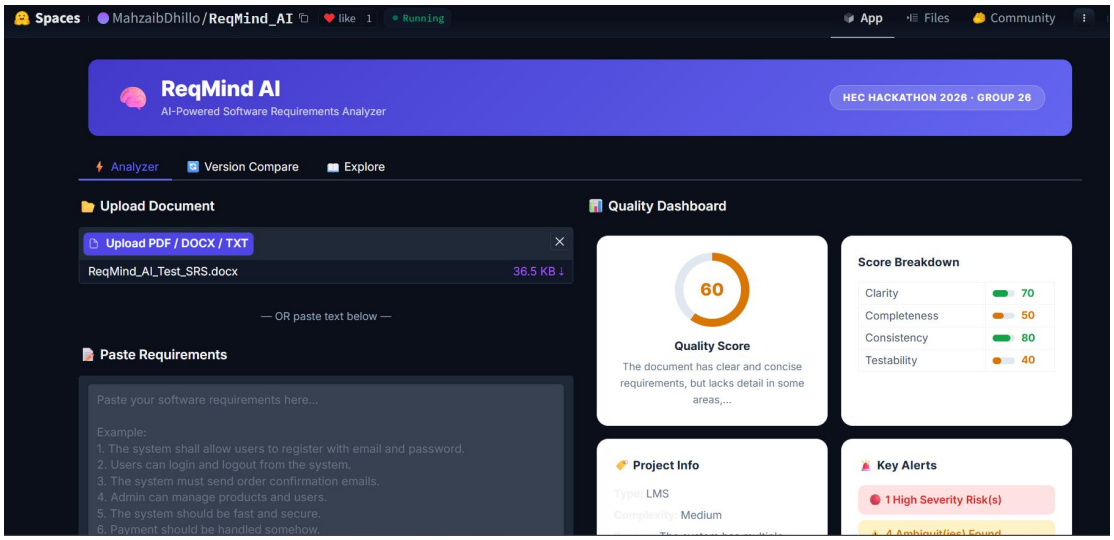
All test inputs were designed to simulate real-world requirement engineering scenarios including ambiguity, multilingual context, scalability demands, and API failure conditions.

Testing case 1



Testing case 2


Spaces
MahzaibDhillo/ReqMind_AI
like 1
Running
App
Files
Community
ReqMind AI
AI-Powered Software Requirements Analyzer
HEC HACKATHON 2026 · GROUP 26
Analyzer
Version Compare
Explore
Upload Document
Upload PDF / DOCX / TXT
Drop File Here
- or -
Click to Upload
— OR paste text below —
Paste Requirements
The system must allow users to register, login, recover password, view products, add products to cart, calculate total price, support 3 payment methods, send confirmation email, scale for 10,000 users, and secure data at all times.
Analyze
Analysis Complete! | Type: E-commerce | Score: 60/100 | Complexity: Medium | Overall: Fair
Quality Dashboard
60
Quality Score
The document is clear and concise, but lacks specific details on some features a...
Score Breakdown
Clarity 70
Completeness 50
Consistency 80
Testability 40
Project Info
Type: E-commerce
Complexity: Medium
Reason: Multiple features and scalability requirement
Key Alerts
1 High Severity Risk(s)
2 Ambiguit(ies) Found
1 Scope Creep Warning(s)
Full JSON
Download PDF
{
"project_info": {
"detected_type": "E-commerce",
"complexity": "Medium",
"complexity_reason": "Multiple features and scalability requirement",
"total_requirements_count": 18
},
"quality_score": {
"overall": 60,
"clarity": 70,
"completeness": 50,
"consistency": 80,
"testability": 40,
"breakdown":
"The document is clear and concise, but lacks specific details on some features and non-functional requirements."
},
"functional_requirements": [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Analyzer
Version Compare
Explore
Upload Document
Upload PDF / DOCX / TXT
ReqMind_AI_Test_SRS.docx
36.5 KB
— OR paste text below —
Paste Requirements
Paste your software requirements here...
Example:
1. The system shall allow users to register with email and password.
2. Users can login and logout from the system.
3. The system must send order confirmation emails.
4. Admin can manage products and users.
5. The system should be fast and secure.
6. Payment should be handled somehow.
Quality Dashboard
60
Quality Score
The document has clear and concise requirements, but lacks detail in some areas,...
Score Breakdown
Clarity 70
Completeness 50
Consistency 80
Testability 40
Project Info
Type: LMS
Complexity: Medium
Key Alerts
1 High Severity Risk(s)